

Donghwan Rho

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Research Interests

- Implementing Language Models Under Homomorphic Encryption
- Mathematical Reasoning in Large Language Models
- Interpretability of Language Models

Education

- Ph.D. Seoul National University**, Mathematical Sciences Mar. 2021 – Present
- Advisor: [Ernest K. Ryu](#)
 - Expected Graduation: Feb. 2026
- BS Korea University**, Mathematics Education Mar. 2015 – Feb. 2021
- Graduated 1st in Class
 - (Mandatory) Military Service: May 2017 - Jan. 2019
 - Teacher's Certificate for Secondary Regular Teacher of Mathematics.

Experience

- Workshop Coordinator**, SNU-KAIST ML Theory Workshop Gangneung, South Korea
Aug. 2024
- Assisted in planning the schedule, selecting the venue, and managing workshop programs.
- Teaching Practicum** Seoul, South Korea
Apr. 2020 - May 2020
- Danggok high school.
- Military Service**, Auxiliary Police Seoul, South Korea
May 2017 – Jan. 2019
- Assisted in maintaining public order and security.

Publications

*: Equal Contribution

- [2] **Studying the Korean Word-Chain Game with RLVR: Mitigating Reward Conflicts via Curriculum Learning** Oct. 2025
Donghwan Rho
 Submitted
[Paper](#)
- [1] **Traveling Salesman-Based Token Ordering Improves Stability in Homomorphically Encrypted Language Models** Oct. 2025
Donghwan Rho*, Sieun Seo*, Hyewon Sung, Chohong Min, Ernest K. Ryu
 Submitted
[Paper](#)
- [0] **Encryption-Friendly LLM Architecture** Oct. 2024
Donghwan Rho*, Taeseong Kim*, Minje Park, Jung Woo Kim, Hyunsik Chae, Ernest K. Ryu, Jung Hee Cheon
 International Conference on Learning Representations (ICLR) 2025
[Paper](#) [Code](#)

Talks & Presentations

- | | |
|--|-----------|
| Seoul National University Research Institute of Mathematics Symposium , Talk | Nov. 2025 |
| • Title: Implementing Efficient Language Models under Homomorphic Encryption | |
| Korean Mathematical Society Annual Meeting , Talk | Oct. 2025 |
| • Title: Traveling Salesman-Based Token Ordering Improves Stability in Homomorphically Encrypted Language Models | |
| SNU-Samsung Electronics Collaboration Symposium , Poster Presentation | Aug. 2024 |
| • Title: Encryption-Friendly LLM Architecture | |
| SNU-KAIST ML Theory Workshop , Poster Presentation | Aug. 2024 |
| • Title: Encryption-Friendly LLM Architecture | |

Honors and Awards

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|---|-----------|
| Outstanding TA , Seoul National University | July 2025 |
| Introduction to Computer Programming and Artificial Intelligence for Scientists | |
| University Students Contest of Mathematics , Silver Prize | Dec. 2018 |
| Academic Excellence Scholarship , Korea University | Aug. 2015 |

Teaching

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|---|---------------------|
| Peer Tutoring in Basic Sciences , Lead Teaching Assistant | Mar. 2024 - Present |
| • Represented section TAs to ensure the smooth operation of the peer tutoring program. | |
| • Oversaw tutorings in Mathematics, Physics, Chemistry, Biology, and Statistics. | |
| Basic Calculus , Teaching Assistant | Sep. 2021 - Present |
| • Matched tutors and tutees for the <i>Basic Calculus</i> peer tutoring program. | |
| • Managed program logistics to ensure efficient operation. | |
| SNU Pre-College Program , Head TA, Instructor | |
| • Managed logistics of the mathematics program for freshman undergraduate. | |
| • Gave lectures. | |
| • Winter 2024, Winter 2023, Winter 2022 | |
| Introduction to Computer Programming and Artificial Intelligence for Scientists , Instructor | Spring 2025 |
| • English Course | |
| • Selected as an outstanding TA  | |
| Topics in Machine Intelligence | Fall 2024 |
| • Topics: Continuous-depth neural networks, Diffusion Models, Large Language Models, Vision Language Models, State-space Models | |
| • English Course | |
| • Graduate Course | |
| Mathematical algorithms II | Fall 2022 |
| • Topics: Reinforcement Learning, Diffusion Models, Natural Language Processing | |
| • English Course | |

- Graduate Course

Calculus 2 Practice, Instructor

- Fall 2021, Fall 2025

Calculus 1 Practice, Instructor

- Spring 2024, Fall 2023, Spring 2023, Spring 2022, Spring 2021

Technical Skills

Languages: Korean (Native), English (Fluent)

Programming Languages: Python, C++

Tools & Technologies: LaTeX, GitHub